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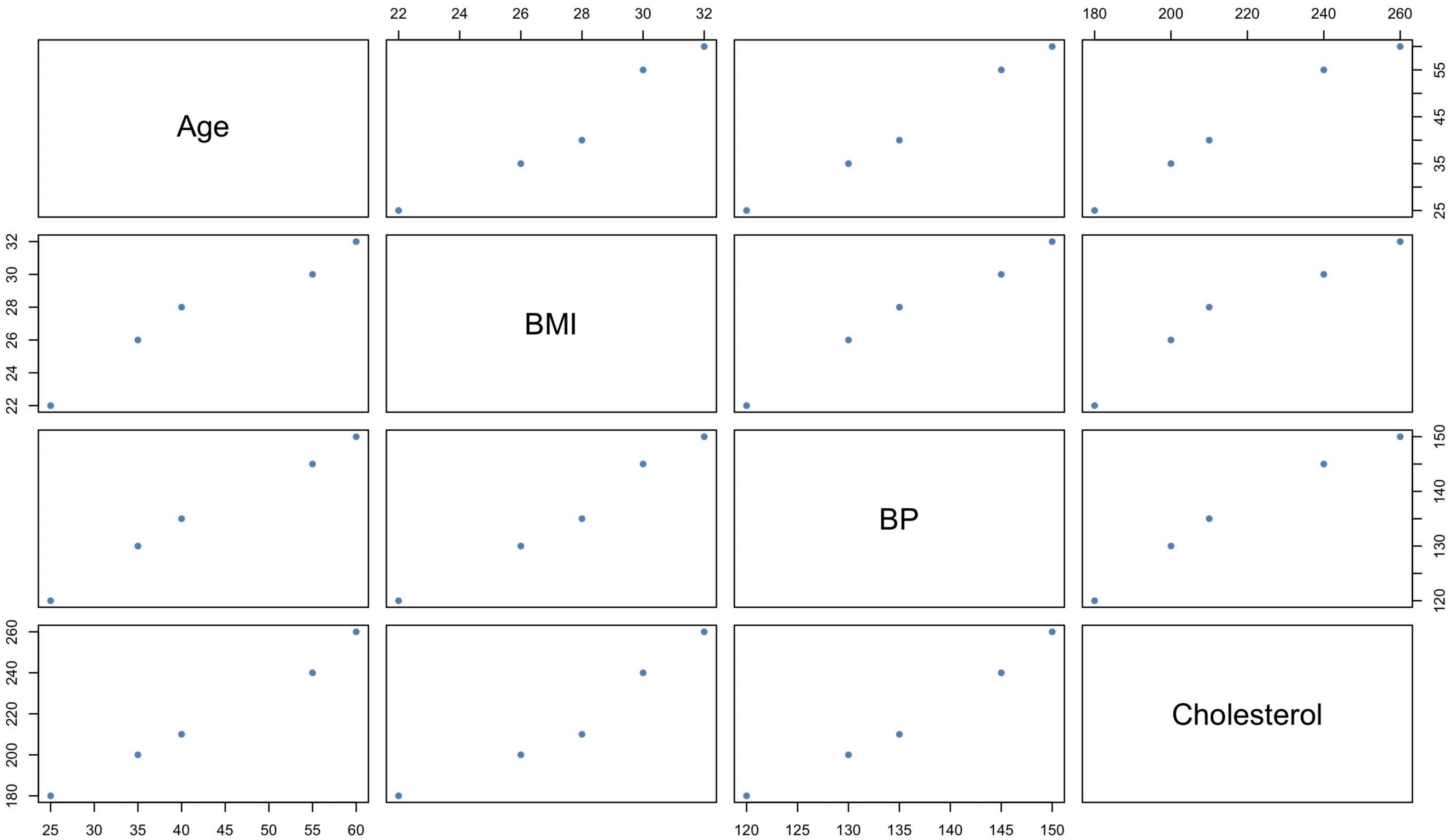
Qv Help Search

```
> # -----
> # Dataset Setup
> # -----
> patient_data <- data.frame(
+   Patient_ID = c("P1","P2","P3","P4","P5"),
+   Age = c(25, 40, 55, 35, 60),
+   BMI = c(22, 28, 30, 26, 32),
+   BP = c(120, 135, 145, 130, 150),
+   Cholesterol = c(180, 210, 240, 200, 260)
+ )
>
> print(patient_data)
  Patient_ID Age  BMI  BP Cholesterol
1         P1  25  22 120          180
2         P2  40  28 135          210
3         P3  55  30 145          240
4         P4  35  26 130          200
5         P5  60  32 150          260
>
> new_window <- function() {
+   if (.Platform$OS.type == "windows") {
+     windows()
+   } else {
+     quartz()
+   }
+ }
>
> # -----
> # Q1: Scatterplot Matrix
> # -----
> new_window()
> pairs(patient_data[, c("Age","BMI","BP","Cholesterol")],
+       main = "Scatterplot Matrix - Patient Health Indicators",
+       col = "steelblue",
+       pch = 16,
+       labels = c("Age","BMI","BP","Cholesterol"))
>
> # -----
> # Q2a: Q-Q Plot for Cholesterol
> # -----
> new_window()
> qqnorm(patient_data$Cholesterol,
+       main = "Q-Q Plot of Cholesterol Levels",
+       col = "darkred",
+       pch = 16)
> qqline(patient_data$Cholesterol, col = "blue", lwd = 2)
>
> # -----
> # Q2b: ECDF for Cholesterol
> # -----
> new_window()
> plot(ecdf(patient_data$Cholesterol),
+       main = "ECDF of Cholesterol Levels",
+       xlab = "Cholesterol")
```

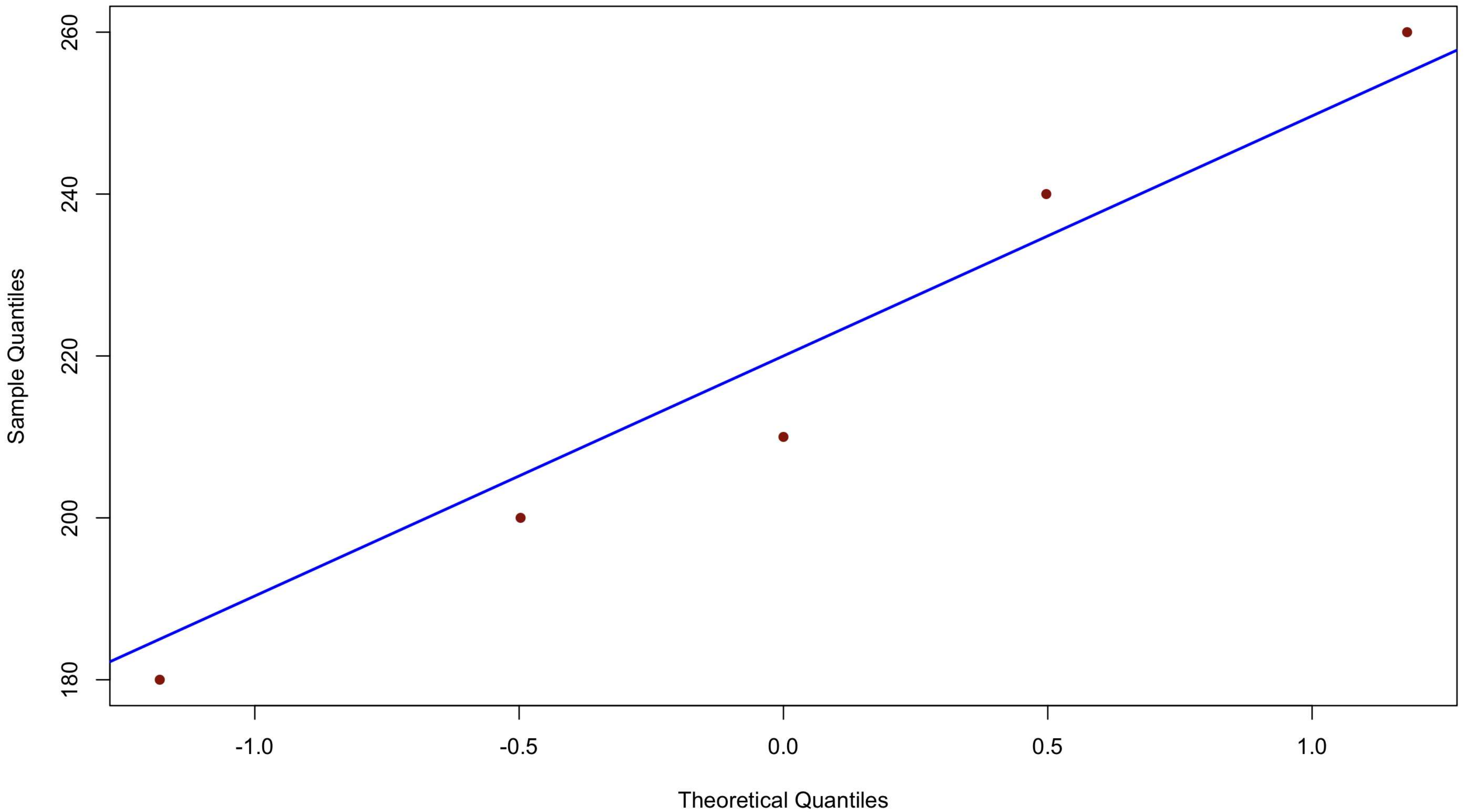


```
> # -----
> new_window()
> pairs(patient_data[, c("Age","BMI","BP","Cholesterol")],
+       main = "Scatterplot Matrix - Patient Health Indicators",
+       col = "steelblue",
+       pch = 16,
+       labels = c("Age","BMI","BP","Cholesterol"))
>
> # -----
> # Q2a: Q-Q Plot for Cholesterol
> # -----
> new_window()
> qqnorm(patient_data$Cholesterol,
+       main = "Q-Q Plot of Cholesterol Levels",
+       col = "darkred",
+       pch = 16)
> qqline(patient_data$Cholesterol, col = "blue", lwd = 2)
>
> # -----
> # Q2b: ECDF for Cholesterol
> # -----
> new_window()
> plot(ecdf(patient_data$Cholesterol),
+     main = "ECDF of Cholesterol Levels",
+     xlab = "Cholesterol",
+     ylab = "Cumulative Probability",
+     col = "darkgreen",
+     lwd = 2)
>
> # -----
> # Q3: Bar Chart - Average Health Indicators
> # -----
> avg_values <- colMeans(patient_data[, c("Age","BMI","BP","Cholesterol")])
> print(avg_values)
      Age      BMI      BP Cholesterol
43.0    27.6   136.0    218.0
>
> new_window()
> barplot(avg_values,
+     col = c("skyblue","orange","lightgreen","salmon"),
+     main = "Average Values of Health Indicators",
+     xlab = "Health Indicator",
+     ylab = "Average Value",
+     ylim = c(0, max(avg_values) + 30),
+     names.arg = c("Age","BMI","BP","Cholesterol"),
+     border = "black")
>
> # Add value labels on top of each bar
> text(x = c(0.7, 1.9, 3.1, 4.3),
+     y = avg_values + 5,
+     labels = round(avg_values, 1),
+     cex = 0.9,
+     font = 2)
>
```

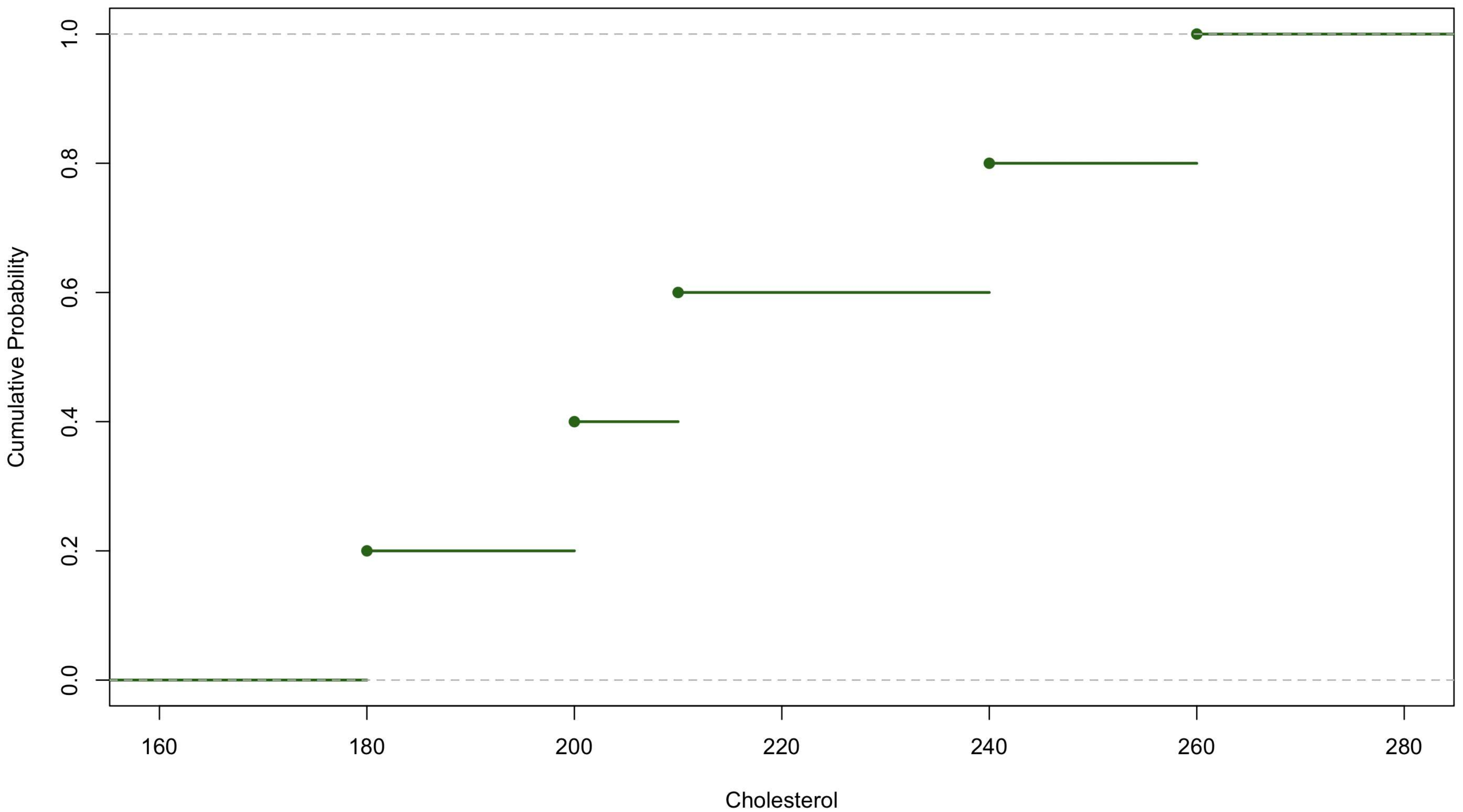
Scatterplot Matrix - Patient Health Indicators



Q-Q Plot of Cholesterol Levels



ECDF of Cholesterol Levels



Average Values of Health Indicators

